Skip to main content

> q-fin > arXiv:1907.11053

Search... GO

quick links

- Login
- Help Pages
- About

Quantitative Finance > Trading and Market Microstructure

arXiv:1907.11053 (q-fin)

[Submitted on 25 Jul 2019 (v1), last revised 7 Mar 2021 (this version, v2)]

Optimal make take fees in a multi market maker environment

Bastien Baldacci, Dylan Possamaï, Mathieu Rosenbaum

View PDF

Following the recent literature on make take fees policies, we consider an exchange wishing to set a suitable contract with several market makers in order to improve trading quality on its platform. To do so, we use a principal-agent approach, where the agents (the market makers) optimise their quotes in a Nash equilibrium fashion, providing best response to the contract proposed by the principal (the exchange). This contract aims at attracting liquidity on the platform. This is because the wealth of the exchange depends on the arrival of market orders, which is driven by the spread of market makers. We compute the optimal contract in quasi explicit form and also derive the optimal spread policies for the market makers. Several new phenomena appears in this multi market maker setting. In particular we show

that it is not necessarily optimal to have a large number of market makers in the presence of a contracting scheme.

Subjects: **Trading and Market Microstructure (q-fin.TR)**; Mathematical Finance (q-fin.MF)
Cite as: **arXiv:1907.11053 [q-fin.TR]**
(or **arXiv:1907.11053v2 [q-fin.TR]** for this version)
https://doi.org/10.48550/arXiv.1907.11053
Focus to learn more
arXiv-issued DOI via DataCite

Submission history

From: Bastien Baldacci [view email]
[v1] Thu, 25 Jul 2019 13:49:51 UTC (474 KB)
[v2] Sun, 7 Mar 2021 16:44:07 UTC (569 KB)

Bibliographic Tools

Bibliographic and Citation Tools

Bibliographic Explorer Toggle
Bibliographic Explorer *(What is the Explorer?)*
Connected Papers Toggle
Connected Papers *(What is Connected Papers?)*
Litmaps Toggle
Litmaps *(What is Litmaps?)*
scite.ai Toggle
scite Smart Citations *(What are Smart Citations?)*

Code, Data, Media

Code, Data and Media Associated with this Article

alphaXiv Toggle
alphaXiv *(What is alphaXiv?)*
Links to Code Toggle
CatalyzeX Code Finder for Papers *(What is CatalyzeX?)*
DagsHub Toggle
DagsHub *(What is DagsHub?)*
GotitPub Toggle
Gotit.pub *(What is GotitPub?)*
Huggingface Toggle
Hugging Face *(What is Huggingface?)*
Links to Code Toggle
Papers with Code *(What is Papers with Code?)*
ScienceCast Toggle
ScienceCast *(What is ScienceCast?)*

○ Demos

Demos

☐ Replicate Toggle
Replicate *(What is Replicate?)*
☐ Spaces Toggle
Hugging Face Spaces *(What is Spaces?)*
☐ Spaces Toggle
TXYZ.AI *(What is TXYZ.AI?)*
○ Related Papers

Recommenders and Search Tools

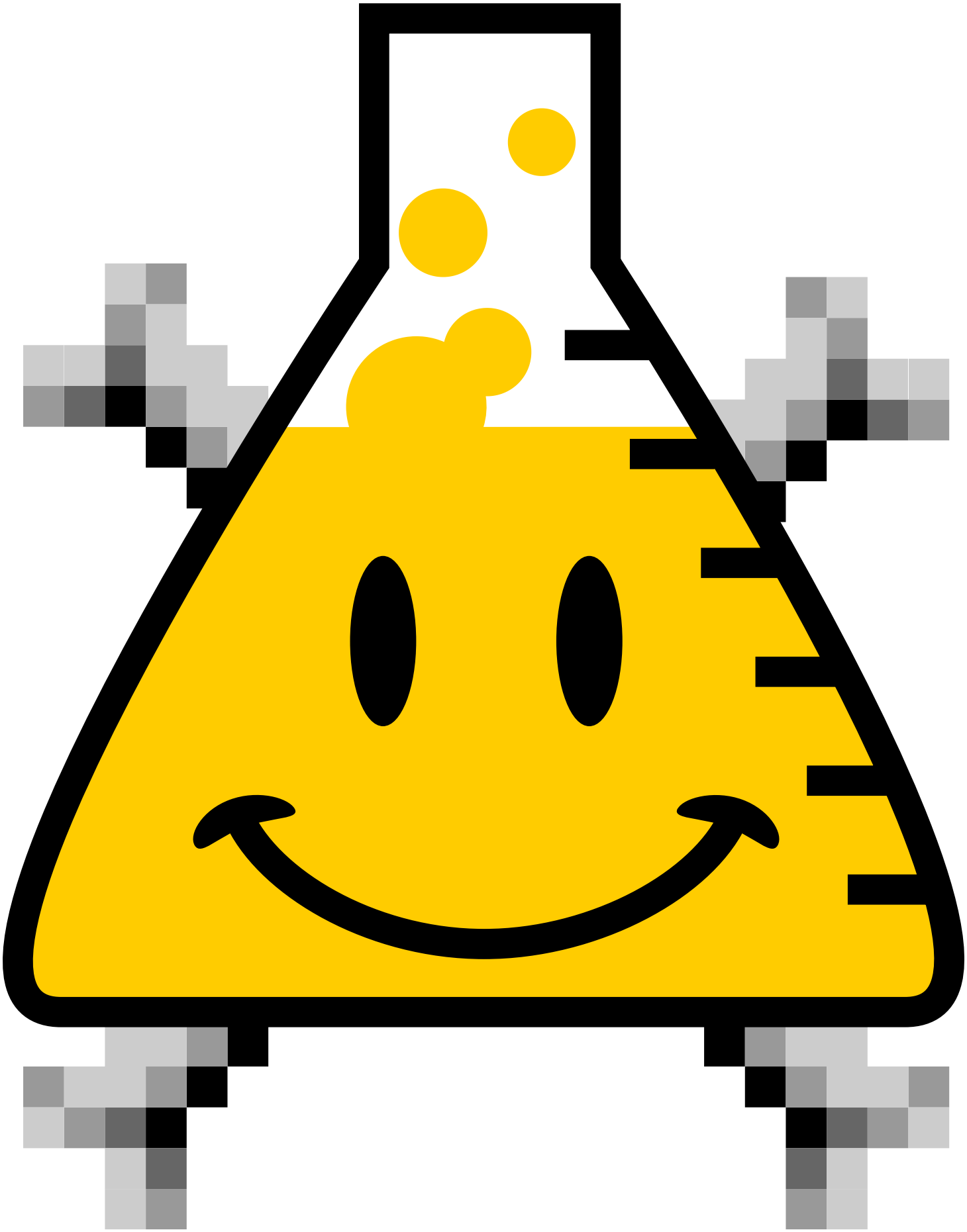
☐ Link to Influence Flower
Influence Flower *(What are Influence Flowers?)*
☐ Core recommender toggle
CORE Recommender *(What is CORE?)*
○ About arXivLabs

arXivLabs: experimental projects with community collaborators

arXivLabs is a framework that allows collaborators to develop and share new arXiv features directly on our website.

Both individuals and organizations that work with arXivLabs have embraced and accepted our values of openness, community, excellence, and user data privacy. arXiv is committed to these values and only works with partners that adhere to them.

Have an idea for a project that will add value for arXiv's community? **Learn more about arXivLabs**.



Which authors of this paper are endorsers? | Disable MathJax (What is MathJax?)